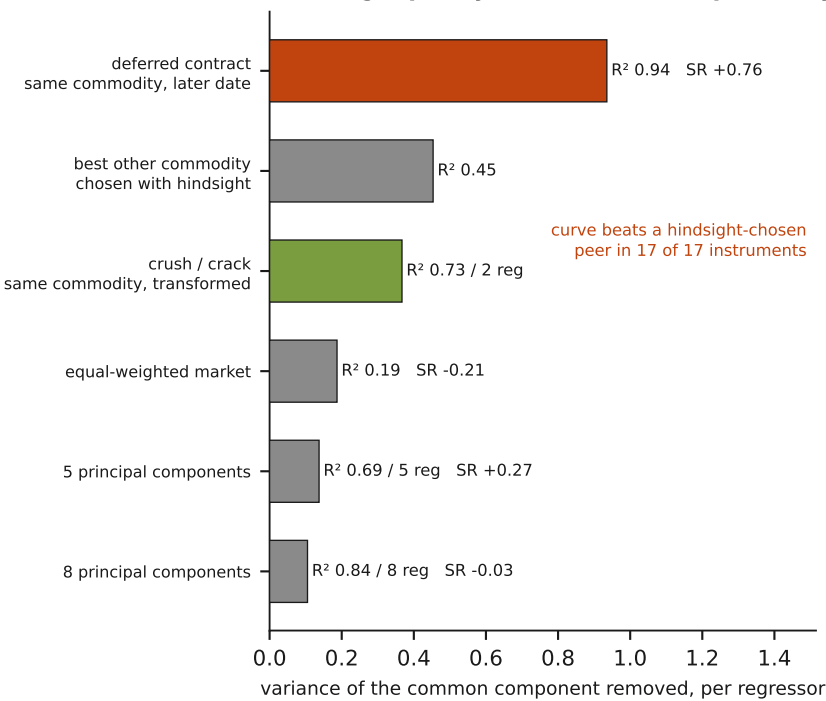


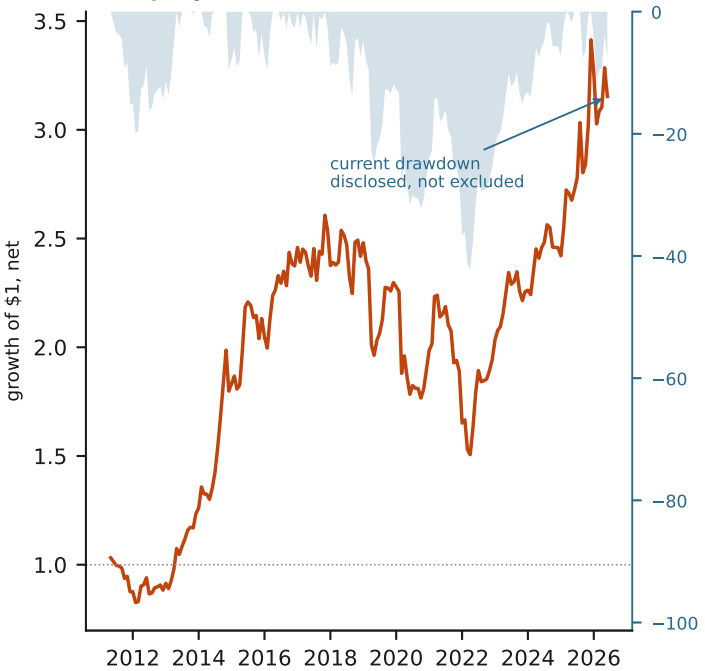
The Same Barrel — Curve-Residual Momentum in Commodity Futures

Supporting exhibits

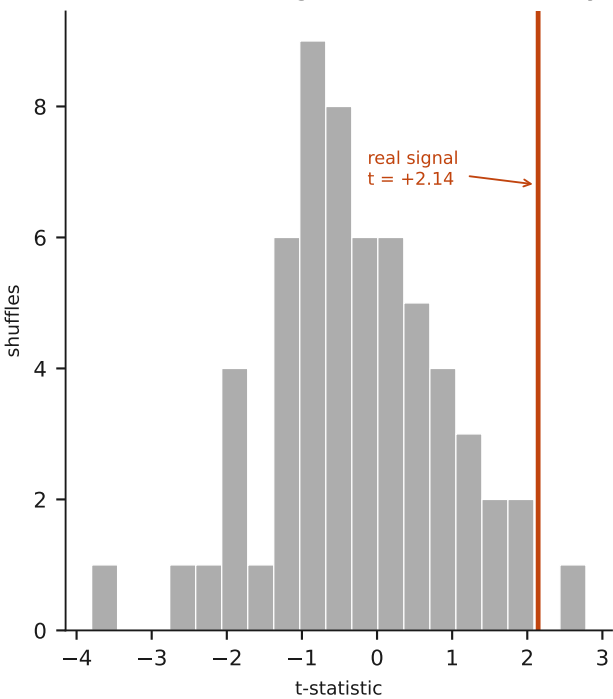
1 • Hedge quality tracks economic proximity



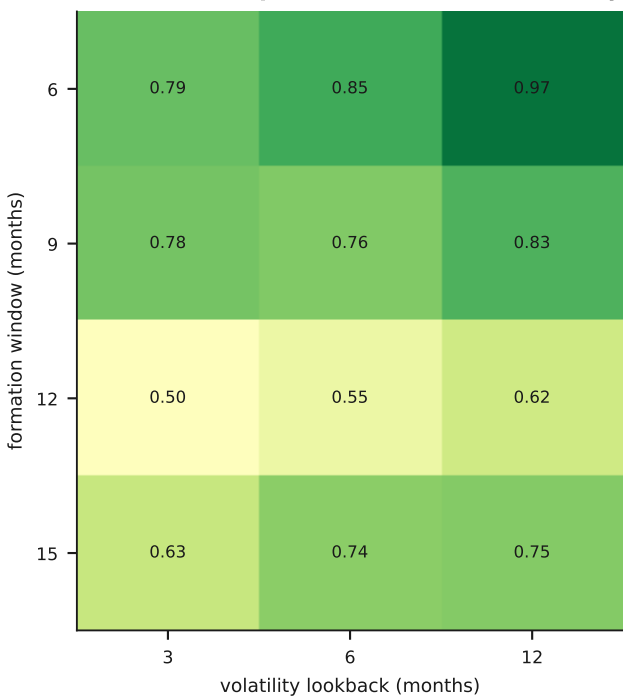
2 • Equity curve and drawdown, 2011-2026



3 • Placebo: the signal, not the machinery



4 • Parameter plateau, and cost sensitivity



60 shuffles of the signal across instruments; all other mechanics fixed
17 CME commodity futures, 2010-06 to 2026-07, 182 monthly observations. All figures net of 3bp per side unless stated. Positions are whole contracts on \$450,000.
Exhibit 1: proximity tiers computed from daily returns; principal-component and market tiers from the channel comparison. Sharpe shown where measured.

net Sharpe by cost per side: 3bp 0.55 10bp 0.50 20bp 0.43 40bp 0.30